

101. If $y = y(x)$ is the solution of the differential equation $\frac{dy}{dx} + \frac{4x}{(x^2-1)}y = \frac{x+2}{(x^2-1)^{\frac{5}{2}}}$, $x > 1$ such that

$y(2) = \frac{2}{9} \log_e(2 + \sqrt{3})$ and $y(\sqrt{2}) = \alpha \log_e(\sqrt{\alpha} + \beta) + \beta - \sqrt{\gamma}$, $\alpha, \beta, \gamma \in \mathbb{N}$, then $\alpha\beta\gamma$ is equal to _____.

यदि $\frac{dy}{dx} + \frac{4x}{(x^2-1)}y = \frac{x+2}{(x^2-1)^{\frac{5}{2}}}$, $x > 1$ का हल $y = y(x)$ है तथा $y(2) = \frac{2}{9} \log_e(2 + \sqrt{3})$ और

$y(\sqrt{2}) = \alpha \log_e(\sqrt{\alpha} + \beta) + \beta - \sqrt{\gamma}$, $\alpha, \beta, \gamma \in \mathbb{N}$ हैं, तो $\alpha\beta\gamma$ बराबर है _____

Ans. 6

Sol. $\frac{dy}{dx} + \frac{4x}{(x^2-1)}y = \frac{x+2}{(x^2-1)^{\frac{5}{2}}}$, $x > 1$ which is linear in y

$$\text{I.f. } e^{\int \frac{4x}{x^2-1} dx} = e^{2 \ln(x^2-1)} = (x^2-1)^2$$

$$\text{Solution is } y(x^2-1)^2 = \int \frac{x+2}{(x^2-1)^{\frac{5}{2}}} (x^2-1)^2 dx + c$$

$$\Rightarrow y(x^2-1)^2 = \int \frac{x+2}{\sqrt{x^2-1}} dx + c$$

$$\Rightarrow y(x^2-1)^2 = \sqrt{x^2-1} + 2 \ln(x + \sqrt{x^2-1}) + c$$

$$\therefore y(2) = \frac{2}{9} \ln(2 + \sqrt{3})$$

$$\therefore \frac{2}{9} \ln(2 + \sqrt{3}) \cdot 9 = \sqrt{3} + 2 \ln(2 + \sqrt{3}) + c$$

$$c = -\sqrt{3}$$

$$\therefore y(x^2-1)^2 = \sqrt{x^2-1} + 2 \ln(x + \sqrt{x^2-1}) - \sqrt{3}$$

$$\text{Put } x = \sqrt{2}$$

$$\therefore y(\sqrt{2}) \cdot 1 = 1 + 2 \ln(\sqrt{2} + 1) - \sqrt{3}$$

$$y(\sqrt{2}) = 2 \ln(\sqrt{2} + 1) + 1 - \sqrt{3}$$

$$= \alpha \ln(\sqrt{\alpha} + \beta) + \beta - \sqrt{\gamma}$$

$$\alpha = 2, \beta = 1, \gamma = 3$$

$$\therefore \alpha \cdot \beta \cdot \gamma = 6$$

102. Let $y = y(x)$ be the solution of the differential equation $xdy - ydx = \sqrt{(x^2 - y^2)}dx, x \geq 1$ with $y(1) = 0$. If the area bounded by the line $x = 1, x = e^\pi, y = 0$ and $y = y(x)$ is $\alpha e^{2\pi} + \beta$, then the value of $10(\alpha + \beta)$ is equal to _____.

माना अवकल समीकरण $xdy - ydx = \sqrt{(x^2 - y^2)}dx, x \geq 1$ का हल $y = y(x)$ है। तथा $y(1) = 0$ है। यदि रेखाओं $x = 1, x = e^\pi, y = 0$ तथा $y = y(x)$ द्वारा घिरे क्षेत्र का क्षेत्रफल $\alpha e^{2\pi} + \beta$ है, तो $10(\alpha + \beta)$ का मान बराबर है _____।

Ans. 4

Sol. $xdy - ydx = \sqrt{x^2 - y^2} dx$

$$\frac{xdy - ydx}{x^2} = \frac{1}{x} \sqrt{1 - \frac{y^2}{x^2}} dx$$

$$\int \frac{d\left(\frac{y}{x}\right)}{\sqrt{1 - \frac{y^2}{x^2}}} = \int \frac{dx}{x}$$

$$\sin^{-1}\left(\frac{y}{x}\right) = \ln|x| + c$$

$$x = 1, y = 0 \Rightarrow c = 0$$

$$\frac{y}{x} = \sin(\ln x)$$

$$y = x \sin(\ln x)$$

$$A = \int_1^{e^\pi} x \sin(\ln x) dx$$

$$x = e^t$$

$$dx = e^t dt$$

$$A = \int_0^\pi e^t \sin(\ln e^t) \times e^t dt$$

$$A = \int_0^{\pi} e^{2t} \sin t \, dt$$

$$A = \frac{e^{2t}}{5} (2 \sin t - \cos t) \Big|_0^{\pi}$$

$$= \frac{e^{2\pi}}{5} (2 \sin \pi - \cos \pi) - \frac{1}{5} (2 \sin 0 - \cos 0)$$

$$A = \frac{e^{2\pi}}{5} (0 + 1) - \frac{1}{5} (-1)$$

$$A = \frac{e^{2\pi}}{5} + \frac{1}{5} = \frac{1}{5} (e^{2\pi} + 1) = \alpha e^{2\pi} + \beta$$

$$\alpha = \frac{1}{5} \quad \beta = \frac{1}{5}$$

$$10 (\alpha + \beta) = 10 \left(\frac{1}{5} + \frac{1}{5} \right) = 4$$

103. If $f(x) = \int \frac{5x^8 + 7x^6}{(x^2 + 1 + 2x^7)^2} dx$, ($x \geq 0$), $f(0) = 0$ and $f(1) = \frac{1}{K}$, then the value of K is :

यदि $f(x) = \int \frac{5x^8 + 7x^6}{(x^2 + 1 + 2x^7)^2} dx$, ($x \geq 0$), $f(0) = 0$ तथा $f(1) = \frac{1}{K}$ है, तो K का मान है _____।

Ans. 4

Sol. $f(x) = \int \frac{5x^8 + 7x^6}{(x^2 + 1 + 2x^7)^2} dx$

$$= \int \frac{\frac{5}{x^6} + \frac{7}{x^8}}{\left(\frac{1}{x^5} + \frac{1}{x^7} + 2 \right)^2} dx$$

$$\text{Let } \frac{1}{x^5} + \frac{1}{x^7} + 2 = t$$

$$\Rightarrow \left(-\frac{5}{x^6} - \frac{7}{x^8} \right) dx = dt$$

$$\Rightarrow f(x) = -\int \frac{dt}{t^2} = \frac{1}{t} + c$$

$$f(x) = \frac{1}{\frac{1}{x^5} + \frac{1}{x^7} + 2} + c$$

$$\Rightarrow f(x) = \frac{x^7}{x^2 + 1 + 2x^7} + c$$

$$f(0) = 0 \Rightarrow c = 0$$

$$f(1) = \frac{1}{4} = \frac{1}{k}$$

$$\Rightarrow k = 4$$

104. Let the tangent at any point P on a curve passing through the points $(1, 1)$ and $\left(\frac{1}{10}, 100\right)$, intersect positive x-axis and y-axis at the points A and B respectively. If $PA : PB = 1 : k$ and $y = y(x)$ is the solution of the differential equation $e^{\frac{dy}{dx}} = kx + \frac{k}{2}$, $y(0) = k$, then $4y(1) - 5\log_e 3$ is equal to _____.

माना बिंदुओं $(1, 1)$ तथा $\left(\frac{1}{10}, 100\right)$ से होकर जाने वाले एक वक्र के किसी बिंदु P पर स्पर्श रेखा धनात्मक x-अक्ष तथा y-अक्षों को

क्रमशः बिंदुओं A तथा B पर काटती है। यदि $PA : PB = 1 : k$ है तथा अवकल समीकरण $e^{\frac{dy}{dx}} = kx + \frac{k}{2}$, $y(0) = k$ का हल $y =$

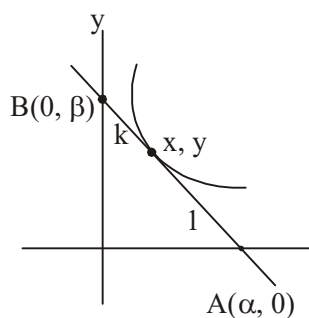
$y(x)$ है, तो $4y(1) - 5\log_e 3$ बराबर है :

Ans. Bonus

Sol. $Y - y = \frac{dy}{dx}(X - x)$

$$Y = 0$$

$$X = \frac{-ydx}{dy} + x$$



$$\frac{k\alpha + 0}{k+1} = 0, \quad \alpha = \frac{k+1}{k}x$$

$$\frac{k+1}{k}x = -y \frac{dx}{dy} + x$$

$$x + \frac{x}{k} = -y \frac{dx}{dy} + x$$

$$x \frac{dy}{dx} + ky = 0 \quad \frac{dy}{dx} + \frac{k}{x}y = 0$$

$$y \cdot x^k = C$$

$$C = 1$$

$$100 \cdot \left(\frac{1}{10}\right)^k = 1$$

$$K = 2$$

$$\frac{dy}{dx} = \ln(2x+1)$$

$$y = \frac{(2x+1)}{2} (\ln(2x+1) - 1) + c$$

$$2 = \frac{1}{2}(0-1) + C$$

$$C = 2 + \frac{1}{2} = \frac{5}{2}$$

$$y(1) = \frac{3}{2}(\ln 3 - 1) + \frac{5}{2}$$

$$= \frac{3}{2} \ln 3 + 1$$

$$4y(1) = 6 \ln 3 + 4$$

$$4y(1) - 5 \ln 3 = 4 + \ell \ln 3$$

105. Let $\{x\}$ and $[x]$ denote the fractional part of x and the greatest integer $\leq x$ respectively of a real number x .
If $\int_0^n \{x\} dx$, $\int_0^n [x] dx$ and $10(n^2 - n)$, ($n \in \mathbb{N}$, $n > 1$) are three consecutive terms of a G.P., then n is equal to _____.

माना $\{x\}$ तथा $[x]$ क्रमशः एक वास्तविक संख्या x के भिन्नात्मक भाग तथा महत्तम पूर्णांक $\leq x$, को दर्शाते हैं। यदि

$\int_0^n \{x\} dx, \int_0^n [x] dx$ तथा $10(n^2 - n)$, ($n \in \mathbb{N}, n > 1$) एक गुणोत्तर श्रेणी के तीन क्रमागत पद हैं, तो n का मान है _____।

Ans. 21

Sol. $\int_0^n \{x\} dx = n \int_0^n x dx = n \left(\frac{x^2}{2} \right)_0^n = \frac{n^2}{2}$

$$\int_0^n \{x\} dx = \int_0^n (x - \{x\}) dx = \int_0^n x dx - \int_0^n \{x\} dx = \frac{n^2}{2} - \frac{n^2}{2}$$

now $\frac{n^2}{2}, \frac{n^2 - n}{2}$ and $10(n^2 - n)$ G.P

$$\left(\frac{n^2 - n}{2} \right)^2 = \frac{n^2}{2} \times 10(n^2 - n)$$

$$\Rightarrow n - 20 = 20$$

$$\Rightarrow n = 21$$

106. If the solution curve of the differential equation $(y - 2 \log_e x) dx + (x \log_e x^2) dy = 0, x > 1$ passes through the points $\left(e, \frac{4}{3}\right)$ and (e^4, α) , then α is equal to _____.

यदि अवकल समीकरण $(y - 2 \log_e x) dx + (x \log_e x^2) dy = 0, x > 1$ का हल वक्र बिंदुओं $\left(e, \frac{4}{3}\right)$ तथा (e^4, α) से होकर जाता है, तो α बराबर है _____

Ans. 3

Sol. $\therefore (y - 2 \ln x) dx + (2x \ln x) dy = 0.$

$$2x \ln x \frac{dy}{dx} + y = 2 \ln x$$

$$\frac{dy}{dx} + \frac{y}{2x \ln x} = \frac{1}{x}$$

$$\therefore \text{I.F.} = e^{\int \frac{1}{2x \ln x} dx} = \sqrt{\ln x}$$

$\therefore$ Solution of the equation is :

$$y \cdot \sqrt{\ln x} = \int \frac{\sqrt{\ln x}}{x} dx$$

$$\therefore y \cdot \sqrt{\ln x} = \frac{2}{3} (\ln x)^{\frac{3}{2}} + C \quad \dots (i)$$

$$\therefore \text{eq. (i) passes through point } \left(e, \frac{4}{3} \right).$$

$$\therefore C = \frac{2}{3}$$

$$\therefore y \sqrt{\ln x} = \frac{2}{3} (\ln x)^{\frac{3}{2}} + \frac{2}{3} \quad \dots (ii)$$

This equation passes through point (e^4, α)

$$\therefore \alpha = 3.$$

107. If $\int_{-a}^a (|x| + |x-2|) dx = 22$ ($a > 2$) and $[x]$ denotes the greatest integer $\leq x$, then $\int_a^{-a} (x + [x]) dx$ is equal to _____.

यदि $\int_{-a}^a (|x| + |x-2|) dx = 22$, ($a > 2$) है तथा $[x]$, महत्तम पूर्णांक $\leq x$ को दर्शाता है, तो $\int_a^{-a} (x + [x]) dx$ बराबर है _____।

Ans. 3

Sol. $I = \int_{-a}^a |x| dx + \int_{-a}^a |x-2| dx = 22$

$$2 \int_0^a x dx + \int_{-a}^2 (2-x) dx + \int_2^a (x-2) dx = 22$$

$$a = 3$$

$$\int_a^{-a} (x + [x]) dx = \int_3^{-3} x dx + \int_3^{-3} [x] dx$$

$$= 0 + 3 = 3$$

108. If for $x \geq 0$, $y = y(x)$ is the solution of the differential equation, $(x+1)dy = ((x+1)^2 + y-3)dx$, $y(2) = 0$, then $y(3)$ is equal to _____.

यदि $x \geq 0$ के लिए $y = y(x)$, अवकल समीकरण $(x+1)dy = ((x+1)^2 + y-3)dx$, $y(2) = 0$, का हल है, तो $y(3)$ का मान है _____।

Ans. 3

Sol. $x \geq 0, y = y(x)$
 $(x+1)dy = ((x+1)^2 + y - 3)dx$

$$\frac{dy}{dx} = \frac{(x+1)^2 - 3}{x+1} + \frac{y}{x+1}$$

$$\frac{dy}{dx} - \frac{y}{x+1} = \frac{(x+1)^2 - 3}{x+1}$$

$$I.F = e^{\int -\frac{1}{(x+1)} dx} = e^{-\ln(1+x)} = \frac{1}{1+x}$$

Solution of diff. equation

$$y \cdot \frac{1}{1+x} = \int \frac{(x+1)^2 - 3}{(x+1)} \cdot \left(\frac{1}{1+x} \right) dx + c$$

$$y \left(\frac{1}{1+x} \right) = \int \left(1 - \frac{3}{(1+x)^2} \right) dx + c$$

$$y = x(x+1) + 3 + c(1+x)$$

$$y(2) = 0$$

$$0 = 6 + 3 + c \cdot 3$$

$$\therefore c = -3$$

$$\therefore y = x + x^2 + 3 - 3 - 3x$$

$$y = x^2 - 2x$$

$$y(3) = 9 - 6 = 3$$

109. Let f be a differentiable function defined on $\left[0, \frac{\pi}{2}\right]$ such that $f(x) > 0$ and

$$f(x) + \int_0^x f(t) \sqrt{1 - (\log_e f(t))^2} dt = e, \quad \forall x \in \left[0, \frac{\pi}{2}\right]. \text{ Then } \left(6 \log_e f\left(\frac{\pi}{6}\right)\right)^2 \text{ is equal to } \underline{\hspace{2cm}}.$$

माना $\left[0, \frac{\pi}{2}\right]$ पर परिभाषित एक अवकलनीय फलन f है जिसके लिए $f(x) > 0$ और $f(x) + \int_0^x f(t) \sqrt{1 - (\log_e f(t))^2} dt = e,$

$$\forall x \in \left[0, \frac{\pi}{2}\right] \text{ है। तब } \left(6 \log_e f\left(\frac{\pi}{6}\right)\right)^2 \text{ बराबर है } \underline{\hspace{2cm}}$$

Ans. 27

Sol. $t(x) = \int_0^x f(t) \sqrt{1 - \ln^2 f(t)} dt = e : \forall x \in \left[0, \frac{\pi}{2}\right]$

Differentiate on both side

$$f \cdot (x) + f(x) \sqrt{1 - \tan^2(x)} = 0$$

$$\int \frac{f(x) dx}{f(x) \sqrt{1 - \tan^2(x)}} = - \int dx$$

Let $(n + \alpha) = U$

$$\frac{1}{f(x)} \cdot f'(x) dx = du$$

$$\int \frac{du}{\sqrt{1 - u^2}} = - \int dx$$

$$\sin^{-1}(\tan(f(x))) = -x + c$$

put $x = 0$

$$\sin^{-1}(\tan(f(0))) = 0 + c$$

$$\sin^{-1}(1) = c$$

$$c = \frac{\pi}{2}$$

$$\sin^{-1}(\tan(f(x))) = -x + \frac{\pi}{2}$$

$$\tan(f(x)) = \sin\left(\frac{\pi}{2} - x\right) = \cos x$$

$$f(x) = e^{\cos x}$$

$$f\left(\frac{\pi}{6}\right) = e^{\frac{\sqrt{3}}{2}}$$

$$\text{Now } \left(6 \tan e^{\frac{\sqrt{3}}{2}}\right)^2 = \left(6 \times \frac{\sqrt{3}}{2}\right)^2 = 27$$

110. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable function such that $f'(x) + f(x) = \int_0^2 f(t) dt$. If $f(0) = e^{-2}$, then $2f(0) - f(2)$ is equal to _____.

माना $f : \mathbb{R} \rightarrow \mathbb{R}$ एक अवकलनीय फलन है, जिसके लिए ताकि $f'(x) + f(x) = \int_0^2 f(t) dt$ है। यदि $f(0) = e^{-2}$ है, तो

$2f(0) - f(2)$ का मान है।

Ans. 1

Sol. $\frac{dy}{dx} + y = k$

$$y \cdot e^x = k \cdot e^x + c$$

$$f(0) = e^{-2}$$

$$\Rightarrow c = e^{-2} - k$$

$$\therefore y = k + (e^{-2} - k) e^{-x}$$

$$\text{now } k = \int_0^2 (k + (e^{-2} - k) e^{-x}) dx$$

$$\Rightarrow k = e^{-2} - 1$$

$$\therefore y = (e^{-2} - 1) + e^{-x}$$

$$f(2) = 2e^{-2} - 1, f(0) = e^{-2}$$

$$2f(0) - f(2) = 1$$

111. If the curve, $y = y(x)$ represented by the solution of the differential equation $(2xy^2 - y)dx + xdy = 0$, passes through the intersection of the lines, $2x - 3y = 1$ and $3x + 2y = 8$, then $|y(1)|$ is equal to :

यदि अवकल समीकरण $(2xy^2 - y)dx + xdy = 0$ का हल $y = y(x)$, रेखाओं $2x - 3y = 1$ तथा $3x + 2y = 8$ के प्रतिच्छेदन बिंदु से होकर जाता है, तो $|y(1)|$ बराबर है _____।

Ans. 1

Sol. $2xy^2 dx - y dx + x dy = 0$

$$2x dx - \frac{1}{y} dx + \frac{x}{y^2} dy = 0$$

$$2x dx = \frac{1}{y} dx - \frac{x}{y^2} dy$$

$$\int 2x dx = \int d\left(\frac{x}{y}\right)$$

$$x^2 = \frac{x}{y} + c$$

put P.O.I. of $2x - 3y = 1$ and $3x + 2y = 8$

$c = 2$ i.e. $(2, y)$

$y(1) = -1$

$|y(1)| = 1$

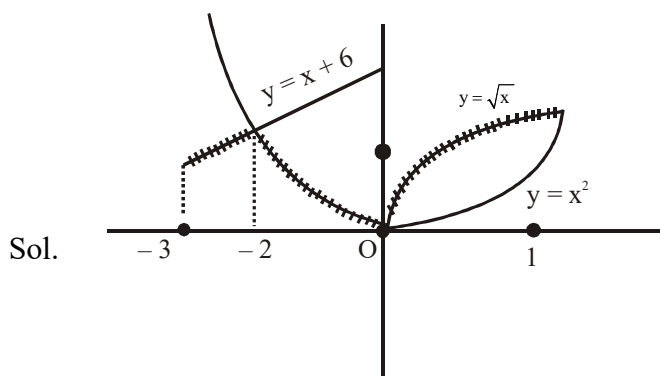
112. Let $f: [-3, 1] \rightarrow \mathbb{R}$ be given as $f(x) = \begin{cases} \min\{(x+6), x^2\} & , -3 \leq x \leq 0 \\ \max\{\sqrt{x}, x^2\} & , 0 \leq x \leq 1 \end{cases}$. If the area bounded by $y = f(x)$ and x-axis is A, then the value of $6A$ is equal to _____.

माना $f: [-3, 1] \rightarrow \mathbb{R}$

$f(x) = \begin{cases} \min\{(x+6), x^2\} & , -3 \leq x \leq 0 \\ \max\{\sqrt{x}, x^2\} & , 0 \leq x \leq 1 \end{cases}$ द्वारा दिया गया है। $y = f(x)$ तथा x-अक्ष द्वारा घिरे क्षेत्र का क्षेत्रफल A है, तो $6A$

बराबर है _____।

Ans. 41



$$\int_{-3}^{-2} (x+6) dx + \int_{-2}^0 x^2 dx + \int_0^1 \sqrt{x} dx$$

$$A = \frac{41}{6}$$

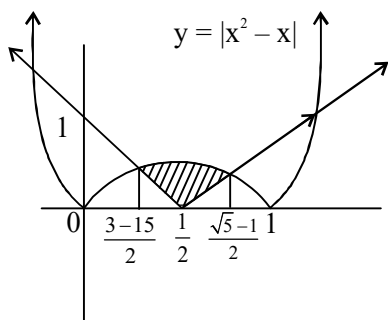
113. Let the area of the region $\{(x, y): |2x-1| \leq y \leq |x^2-x|, 0 \leq x \leq 1\}$ be A. Then $(6A + 11)^2$ is equal to _____.

माना क्षेत्र $\{(x, y): |2x-1| \leq y \leq |x^2-x|, 0 \leq x \leq 1\}$ का क्षेत्रफल A है। तो $(6A + 11)^2$ बराबर है _____

Ans. 125

Sol. $y \geq |2x - 1|$

$$y \leq |x^2 - x|$$



Both curve are symmetric about $x = \frac{1}{2}$ Hence

$$A = 2 \int_{\frac{3-\sqrt{5}}{2}}^{\frac{1}{2}} ((x-x^2) - (1-2x)) dx$$

$$A = 2 \int_{\frac{3-\sqrt{5}}{2}}^{\frac{1}{2}} (-x^2 + 3x - 1) dx = 2 \left(\frac{-x^3}{3} + \frac{3}{2}x^2 - x \right) \Big|_{\frac{3-\sqrt{5}}{2}}^{\frac{1}{2}}$$

on solving

$$6A + 11 = 5\sqrt{5}$$

$$(6A + 11)^2 = 125$$

114. Let the solution curve $x = x(y)$, $0 < y < \frac{\pi}{2}$, of the differential equation

$$(\log_e(\cos y))^2 \cos y dx - (1 + 3x \log_e(\cos y)) \sin y dy = 0 \text{ satisfy } x\left(\frac{\pi}{3}\right) = \frac{1}{2 \log_e 2}. \text{ If}$$

$$x\left(\frac{\pi}{6}\right) = \frac{1}{\log_e m - \log_e n}, \text{ where } m \text{ and } n \text{ are coprime, then } mn \text{ is equal to } \underline{\hspace{2cm}}.$$

माना अवकल समीकरण $(\log_e(\cos y))^2 \cos y dx - (1 + 3x \log_e(\cos y)) \sin y dy = 0$, का हल वक्र $x = x(y)$, $0 < y < \frac{\pi}{2}$;

$x\left(\frac{\pi}{3}\right) = \frac{1}{2 \log_e 2}$ को संतुष्ट करता है। यदि $x\left(\frac{\pi}{6}\right) = \frac{1}{\log_e m - \log_e n}$ है, जहाँ m तथा n असहभाज्य है, तो mn बराबर है :

Ans. 12

Sol. $\cos y \ln^2 \cos y dx = (1 + 3x \ln \cos y) \sin y dy$

$$\frac{dx}{dy} = \tan y \left(\frac{3x}{\ln \cos y} + \frac{1}{\ln^2 \cos y} \right)$$

$$\frac{dx}{dy} - \left(\frac{3 \tan y}{\ln \cos y} \right) x = \frac{\tan y}{\ln^2 \cos y}$$

$$\text{IF} = e^{\frac{-3 \sin y}{\ln \cos y \cos y} dy}$$

$$\ln \cos y = t$$

$$\frac{1}{\cos y} (-\sin y) dy = dt$$

$$\text{If} = e^{\frac{3}{t} dt} = e^{3 \ln t} = t^3 = \ln^3 \cos y$$

$$\text{solution is } x \cdot \ln^3 \cos y = \frac{\sin y}{\cos y} \cdot \ln \cos y dy + C$$

$$x \ln^3 \cos y = \frac{-\ln^2 \cos y}{2} + C$$

$$x \left(\frac{\pi}{3} \right) = \frac{1}{2 \ln 2} \text{ so } \frac{1}{2 \ln 2} \times \ln^3 \left(\frac{1}{2} \right) = -\frac{\ln^2 \left(\frac{1}{2} \right)}{2} + C$$

$$C = 0$$

$$y = \frac{\pi}{6} \Rightarrow x \ln^3 \frac{\sqrt{3}}{2} = -\frac{1}{2} \ln^2 \frac{\sqrt{3}}{2} + 0$$

$$x = -\frac{1}{2 \ln \left(\frac{\sqrt{3}}{2} \right)}$$

$$x = \frac{1}{\ln \frac{4}{3}} = \frac{1}{\ln 4 - \ln 3}$$

$$mn = 12$$

115. Let the normals at all the points on a given curve pass through a fixed point (a, b). If the curve passes through (3, -3) and $(4, -2\sqrt{2})$, and given that $a - 2\sqrt{2}b = 3$, then $(a^2 + b^2 + ab)$ is equal to _____.

माना एक वक्र के प्रत्येक बिंदु पर अभिलम्ब, बिंदु (a, b) से होकर जाते हैं। यदि यह वक्र बिंदुओं (3, -3) तथा $(4, -2\sqrt{2})$ से होकर जाता है, तथा $a - 2\sqrt{2}b = 3$ है, तो $(a^2 + b^2 + ab)$ बराबर है _____।

Ans. 9

Sol. Equation of normal at P(x, y)

$$Y - y = \frac{1}{\frac{dy}{dx}} (X - x)$$

It passes through (a, b)

$$\Rightarrow b - y = - \frac{dx}{dy} (a - x)$$

$$\Rightarrow (b - y) dy = (x - a) dx$$

Integrate both sides

$$\int (b - y) dy = \int (x - a) dx$$

$$\Rightarrow by - \frac{y^2}{2} = \frac{x^2}{2} - ax$$

$$x^2 + y^2 - 2ax - 2by = 0$$

passes through (3, -3) and $(4, -2\sqrt{2})$

$$9 + 9 - 6a + 6b = 0$$

$$\therefore a - b = 3 \quad \dots\dots(i)$$

$$16 + 8 - 8a + 4\sqrt{2}b = 0$$

$$2a - \sqrt{2}b = 6 \quad \dots\dots(ii)$$

$$(i) \times (2) - (ii)$$

$$b(\sqrt{2} - 2) = 0$$

$$b = 0$$

$$\therefore a = 3$$

$$\therefore a^2 + b^2 + ab = 3^2 + 0 + 0$$

$$= 9.$$

116. Let for $x \in \mathbb{R}$ $f(x) = \frac{x + |x|}{2}$ and $g(x) = \begin{cases} x, & x < 0 \\ x^2, & x \geq 0 \end{cases}$.

The area bounded by the curve $y = (f \circ g)(x)$ and the lines $y = 0$, $2y - x = 15$ is equal to _____.

माना $x \in \mathbb{R}$ के लिए $f(x) = \frac{x + |x|}{2}$ तथा $g(x) = \begin{cases} x, & x < 0 \\ x^2, & x \geq 0 \end{cases}$.

है। तो वक्र $y = (f \circ g)(x)$ तथा रेखाओं $y = 0$, $2y - x = 15$ से घिरे क्षेत्र का क्षेत्रफल बराबर है _____

Ans. 72

Sol. $f(x) = \frac{x + |x|}{2} = \begin{cases} x & x \geq 0 \\ 0 & x < 0 \end{cases}$

$$g(x) = \begin{cases} x^2 & x \geq 0 \\ x & x < 0 \end{cases}$$

$$f \circ g(x) = f[g(x)] = \begin{cases} g(x) & g(x) \geq 0 \\ 0 & g(x) < 0 \end{cases}$$

$$f \circ g(x) = \begin{cases} x^2 & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$2y - x = 15$$

$$A = \int_0^3 \left(\frac{x+15}{2} - x^2 \right) dx + \frac{1}{2} \times \frac{15}{2} \times 15$$

$$\left. \frac{x^2}{4} + \frac{15x}{2} - \frac{x^3}{3} \right|_0^3 + \frac{225}{4}$$

$$= \frac{9}{4} + \frac{45}{2} - 9 + \frac{225}{4} = \frac{99 - 36 + 225}{4}$$

$$= \frac{288}{4} = 72$$

